

DSA0204: COMPUTER VISION & OPENCV

Concept Mapping Assessment

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Code of Conduct

I PRASHANTH A K (192324015) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prashanth A K

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Question 13: Analog Scene to Digital Image: Concept Map

The concept map below traces the transformation of a real-world scene into a digital image across three linked domains: the physical domain, where light and optics behave continuously; the sensing/digitization stage, which bridges the physical and computational worlds; and the computational domain, where the scene exists as a discrete, processable data structure.

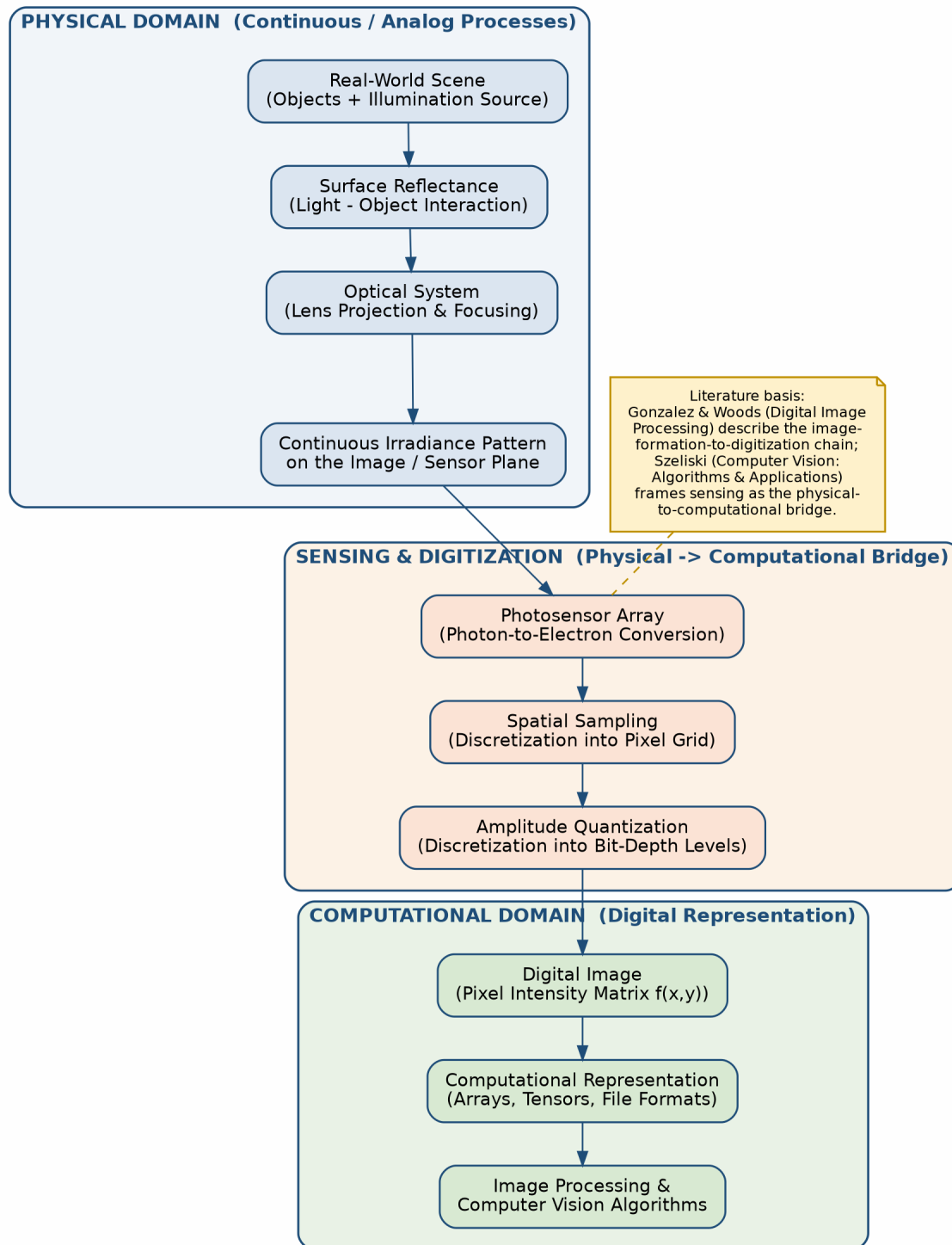


Figure 1. Concept map of the analog-to-digital image transformation pipeline.

Explanation of key linkages

- Scene -> Reflectance -> Optics -> Continuous Signal: models the physical chain by which illumination interacts with object surfaces and is focused by the lens onto the sensor plane as a continuous irradiance pattern (the classical image formation model).
- Continuous Signal -> Sensor -> Sampling -> Quantization: represents the sensing and digitization bridge, where the photosensor array converts photons to electrical signal, spatial sampling discretizes the scene into a pixel grid, and quantization discretizes brightness into a finite set of levels.
- Quantization -> Digital Image -> Data Structures -> CV Algorithms: represents the computational domain, where the digitized result becomes a pixel-intensity matrix that can be stored, represented as arrays/tensors, and consumed by image processing and Computer Vision algorithms.

Literature-based insight: this three-domain structure mirrors the classical image formation and digitization chain described in Gonzalez and Woods' Digital Image Processing, and aligns with Szeliski's framing in Computer Vision: Algorithms and Applications, where image sensing is explicitly treated as the bridge between the physical world and computational representation. Presenting the map in three clearly bounded clusters (physical, transition, computational) keeps the direction of transformation visually unambiguous and easy to follow left-to-right/top-to-bottom.

Question 14: Dependencies Between Image Quality Parameters and System Performance

This concept map identifies how the three core image-quality parameters (resolution, noise, and contrast) each influence system performance through an intermediate mechanism, while also being interdependent with one another rather than fully independent variables.

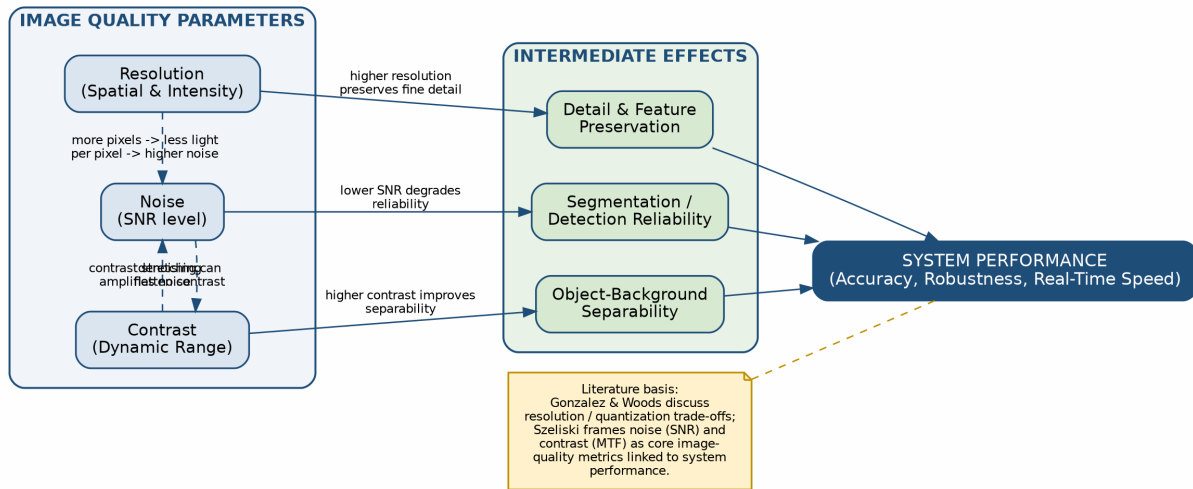


Figure 2. Concept map of image-quality parameter dependencies and their effect on system performance.

Explanation of key linkages

- Resolution -> Detail & Feature Preservation -> System Performance: higher spatial/intensity resolution preserves fine edges and textures that feature-based detection and recognition algorithms rely on.
- Noise -> Segmentation/Detection Reliability -> System Performance: a lower signal-to-noise ratio degrades the reliability of thresholding, segmentation and detection, increasing false positives/negatives.
- Contrast -> Object-Background Separability -> System Performance: stronger contrast improves the separability of objects from background, aiding detection and segmentation accuracy.
- Cross-parameter dependencies (dashed edges): increasing resolution can reduce the light captured per pixel and raise noise; noise-reduction filtering can flatten contrast; and contrast enhancement (stretching) can amplify existing noise. These dashed links show that the three parameters cannot be optimised independently.

Literature-based insight: Gonzalez and Woods discuss the resolution-versus-quantization trade-off and how coarse sampling or quantization degrades usable image information, while Szeliski frames noise (signal-to-noise ratio) and contrast (related to modulation transfer function) as core, interacting image-quality metrics that jointly determine downstream system performance rather than acting in isolation. The map is organised left-to-right (parameters -> mediating effects -> performance) with the interdependency edges drawn separately (dashed) to keep the primary causal structure visually distinct from the secondary trade-off relationships.

Question 15: Image Formation Models and Computer Vision Applications

This concept map links the three theoretical components of the image formation model, geometric, radiometric and sensor models, to the system capabilities they enable, and in turn to the practical Computer Vision applications that depend on those capabilities.

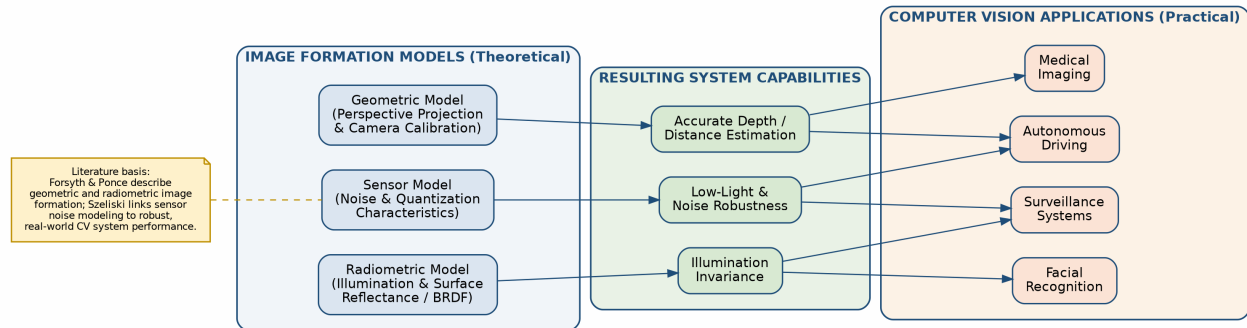


Figure 3. Concept map linking image formation models to system capabilities and practical Computer Vision applications.

Explanation of key linkages

- Geometric Model -> Depth/Distance Estimation -> Autonomous Driving & Medical Imaging: accurate camera calibration and perspective projection modelling enable reliable depth and distance estimation, which is essential for obstacle-distance judgement in autonomous vehicles and for precise spatial measurement in medical scans.
- Radiometric Model -> Illumination Invariance -> Facial Recognition & Surveillance: modelling illumination and surface reflectance allows systems to separate genuine object appearance from lighting variation, improving robustness of facial recognition and surveillance systems across changing lighting conditions.
- Sensor Model -> Low-Light/Noise Robustness -> Autonomous Driving & Surveillance: understanding sensor noise and quantization behaviour allows perception pipelines to remain reliable in low-light or noisy conditions, which is critical for night-time driving and continuous surveillance operation.

Literature-based insight: Forsyth and Ponce's treatment of geometric and radiometric image formation provides the theoretical basis for the left-hand side of the map, while Szeliski's discussion of sensor noise modelling and its role in real-world system robustness supports the sensor-model branch. Grouping the map into three clearly labelled clusters (theoretical models, resulting capabilities, practical applications) keeps the direction of dependency, from theory to capability to application, visually explicit and avoids conflating abstract models with concrete system outcomes.